

AMIRHOSSEIN DABIRIAGHDAM

University of British Columbia, Vancouver, Canada

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RESEARCH INTERESTS

- Large Language Models
- Vision-Language Models
- LLM Agents
- World Models

EDUCATION

University of British Columbia (UBC)

- **Ph.D.** in Electrical and Computer Engineering Sep. 2025 - Present

GPA: 98/100 (A+)

- **M.A.Sc.** in Electrical and Computer Engineering Sep. 2023 - Aug. 2025

Thesis: "Combating Disinformation in the Age of Generative AI: From Watermarking LLMs to Persuasion Analysis of Memes"

GPA: 96.3/100 (A+) (Thesis Grade: 100/100)

University of Tehran (UT)

Sep. 2018 - Jul. 2023

- **B.Sc.** in Electrical and Computer Engineering

Thesis: "An Analysis of Forgettable Examples Extracted During Multilingual Models Training"

GPA: 19.21/20 (Ranked 2nd among 120 students)

SELECTED PUBLICATIONS & PREPRINTS

- SimMark: A Robust Sentence-Level Similarity-Based Watermarking Algorithm for LLMs
Amirhossein Dabiriaghdam, Lele Wang
EMNLP 2025 Main Conference Oral, Suzhou, China, November 2025.
- SOI Matters: Analyzing Multi-Setting Training Dynamics in Pretrained Language Models via Subsets of Interest
Shayan Vassef*, **Amirhossein Dabiriaghdam***, Mohammadreza Bakhtiari*, Yadollah Yaghoobzadeh
Proceedings of the 5th Workshop on Multilingual Representation Learning (MRL 2025) @ EMNLP 2025, Suzhou, China, November 2025.
- BCAmirs at SemEval-2024 Task 4: Beyond Words: A Multimodal and Multilingual Exploration of Persuasion in Memes
*Amirhossein Abaskohi**, **Amirhossein Dabiriaghdam***, Lele Wang, Giuseppe Carenini (*Equal Contribution)
Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024) @ NAACL 2024, Mexico City, Mexico, June 2024.
- Targeted Adversarial Attacks against Neural Machine Translation
Sahar Sadrizadeh, **Amirhossein Dabiriaghdam**, Ljiljana Dolamic, Pascal Frossard
IEEE ICASSP, Rhodes Island, Greece, June 2023.
- When Minor Edits Matter: LLM-Driven Prompt Attack for Medical VLM Robustness in Ultrasound
Yasamin Medghalchi, Milad Yazdani, **Amirhossein Dabiriaghdam**, Moein Heidari, et al.
arXiv preprint arXiv:2603.21047, 2026.
- UnpredictaBench: A Benchmark for Evaluating Distributional Randomness in LLMs
*Amirhossein Abaskohi**, **Amirhossein Dabiriaghdam***, Liang Luo, Ellie Dingqiao Wen, Lele Wang, Giuseppe Carenini, Peter West
arXiv preprint arXiv:2606.06622, 2026. (*Equal Contribution)
- VAMPS: Visual-Assisted Mathematical Problem Solving Benchmark
Amirhossein Dabiriaghdam, Shayan Vassef, Mohammadreza Bakhtiari, Yasamin Medghalchi, Ilker Hacihaliloglu, Mesrob Ohannessian, Lele Wang, Giuseppe Carenini
arXiv preprint arXiv:2606.04244, 2026.

RESEARCH EXPERIENCE

- **Graduate Research Assistant**

University of British Columbia

Sep. 2023 - Present

Vancouver, Canada

- **Ph.D.:** Researching *agentic multimodal LLMs* for robust and interpretable reasoning over interactive visual and decision-support environments, with a focus on *visual tool use*, *world models* for anticipatory reasoning, and *multi-agent collaboration*; developing benchmarks to systematically evaluate model capabilities, failure modes, and generalization.

Under the supervision of Prof. Giuseppe Carenini and Prof. Lele Wang.

- **M.A.Sc.:** Developed *SimMark*, a robust sentence-level similarity-based *watermarking algorithm* for LLM-generated text (EMNLP 2025), and studied multimodal and multilingual persuasion analysis.

- **Research Internship - EPFL Excellence in Engineering (E3)**

Signal Processing Laboratory 4 (LTS4), EPFL

Jul. 2022 - Sep. 2022

Lausanne, Switzerland

- Worked on the targeted *adversarial attacks* against neural machine translation models (ICASSP 2023).
- Under the supervision of Dr. Pascal Frossard.

- **Summer Internship**

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Jul. 2021 - Sep. 2021

Tehran, Iran

- Developed a chatbot *natural language understanding unit* using state-of-the-art deep learning models.
- Under the supervision of Dr. Reshad Hosseini.

HONORS & AWARDS

- Awarded 4 Year Doctoral Fellowship (4YF) from UBC. 2025 - 2029
- Received GSI Award from UBC. 2023 - 2025
- Received the Faculty of Engineering (FOE) Award from the University of Tehran. 2019
- Ranked 537th among about 150000 participants in the Nationwide University Entrance Exam. 2018
- Kyokushin Karate Black Belt holder and member of IKO Kyokushinkaikan. 2008 - Present

RELEVANT COURSES

- | | |
|---|---|
| ● Natural Language Processing (100/100) | ● Machine Learning (92/100) |
| ● NLP Commonsense (99/100) | ● Reinforcement Learning (100/100) |
| ● NLP Meets HCI (90/100) | ● Information and Coding Theory (94/100) |
| ● NLP Open Questions (96/100) | ● Artificial Intelligence (100/100) |
| ● Fundamentals of Visual Computing (93/100) | ● Linear Algebra (100/100) |
| ● Deep Learning with Structures (95/100) | ● Data Structures and Algorithms (98.5/100) |

TEACHING EXPERIENCE

- **Graduate Teaching Assistant**

University of British Columbia

Jan. 2024 - Present

Vancouver, Canada

- AI 422: Intelligent Systems
- AI 322: Foundations of Artificial Intelligence
- CPEN 311: Digital Systems Design

- **Graduate Academic Assistant**

University of British Columbia

Jun. 2026 - Sep. 2026

Vancouver, Canada

- AI 100: Introduction to Artificial Intelligence
Contributed to the design and development of course content, assessments, lab activities, instructional materials, and software infrastructure for a broad-audience introductory AI course. Collaborated with Prof. Kevin Leyton-Brown, Prof. Giulia Toti, and Prof. Elham Khoda, along with other AI 100 faculty.

TECHNICAL SKILLS

- **Programming Languages**
Python, C/C++, MATLAB, Verilog, Visual Basic
Familiar with C#, Java, PHP, SQL, JS, Assembly
 - **ML / AI Frameworks**
Hugging Face Transformers, PyTorch
Familiar with NumPy, Pandas, Scikit-learn, OpenCV
- **Engineering & Simulation**
MATLAB Simulink®, ModelSim, Quartus
Familiar with Proteus, CoppeliaSim, ROS, Gazebo
 - **Technologies & Tools**
MQTT, Git, Makefile, L^AT_EX
Familiar with ARM (STM32), AVR (Arduino), ESP32

LANGUAGES

- **English** Full professional proficiency
 - **French** Elementary proficiency
- **Turkish (Azari)** Native (Bilingual proficiency)
 - **Persian** Native (Bilingual proficiency)

SERVICE & LEADERSHIP

- **Graduate Student Representative** 2025 - 2026
ECE Graduate Student Association (ECEGSA), UBC
 - **Reviewer** 2025
International Symposium on Topics in Coding (ISTC 2025)
 - **Staff Volunteer** 2024
18th Canadian Workshop on Information Theory (CWIT 2024)

REFERENCES

- Dr. Giuseppe Carenini [Full Professor]

More available upon request.
- Dr. Lele Wang [Associate Professor]